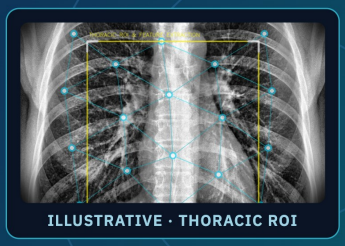
MEDICAL AI · CHEST X-RAY RESEARCH

# Explainable COVID-19 Detection from Chest X-Ray Images Using Deep Learning and Transfer Learning

**Transfer Learning • Explainable AI • Uncertainty Estimation • Robustness Analysis**



ILLUSTRATIVE · THORACIC ROI

| ACCURACY | EXPLAINABILITY | CALIBRATION | ROBUSTNESS |
|---|---|---|---|
| Does it classify reliably? | Where does it look? | Can its confidence be trusted? | Is it stable under perturbations? |

## 1 RESEARCH PROBLEM

Chest X-ray classifiers can learn misleading image patterns, produce poorly calibrated confidence scores and offer little transparency. This work studies classification performance together with explainability, uncertainty and robustness.

## 2 OBJECTIVES

- Develop a three-class chest X-ray classifier.
- Compare transfer-learning architectures.
- Visualize model attention using explainable AI.
- Evaluate confidence calibration and robustness.

## 3 PROPOSED METHODOLOGY

Data · Preprocessing · Model · Evaluation · Explain & demo

1. Dataset Acquisition
2. Annotation Verification
3. Image Quality and Duplicate Checks
4. Preprocessing and Augmentation
5. Patient-Level Split (when identifiers permit)
6. Transfer Learning
7. Evaluation
8. Explainability and Uncertainty Analysis
9. Interactive Demonstration

**DenseNet121**
Dense connectivity: each layer reuses all earlier feature maps

**EfficientNet-B3**
Compound scaling of depth, width and resolution

**ResNet50**
Residual skip connections for deep, stable training

→ COVID-19 · Normal · Pneumonia

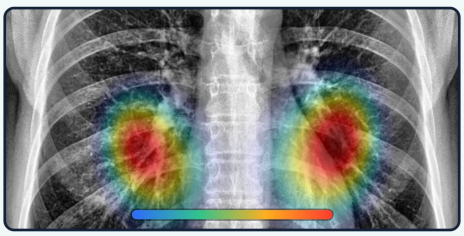
Setup: ImageNet-pretrained · 300 × 300 input · class-weighted loss · head warm-up, then fine-tuning · temperature scaling on validation data

## 4 EXPLAINABLE AI

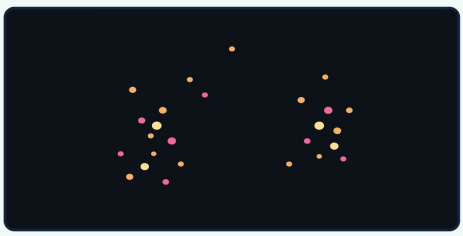
Highlighted regions do not prove the model's reasoning



Original chest X-ray · Illustrative visualization



Grad-CAM heatmap overlay · Illustrative visualization



Integrated Gradients · Illustrative visualization

## 5 EVALUATION FRAMEWORK

Plan only · results pending

| | | | |
|---|---|---|---|
| Macro-F1 and per-class recall — Results pending | Confusion matrix — Results pending | One-vs-rest ROC-AUC — Results pending | COVID-19 sensitivity and specificity — Results pending |
| Calibration: ECE and Brier score — Results pending | Robustness under image perturbations — Results pending | Inference latency — Results pending | Abstention coverage vs accuracy — Results pending |

## 6 UNCERTAINTY-AWARE PREDICTION

Image Input
↓
Predicted Class + Calibrated Confidence
↓
Prediction | "Uncertain — Expert Review Required"

Abstention is a research safeguard and does not guarantee clinical safety.

## 7 EXPECTED CONTRIBUTIONS

- Reproducible comparison of three pretrained architectures
- Visual interpretation of model predictions
- Confidence calibration and uncertainty analysis
- Robustness testing and artifact investigation
- An interactive research demonstration

## 8 LIMITATIONS AND ETHICS

- Dataset quality and representativeness may limit generalization.
- Data leakage and acquisition artifacts must be investigated.
- External clinical validation is required before real-world use.
- This is a research prototype, not a diagnostic device.